

U.G.C. NET EX COMPUTER (Solved with Expla

1. Consider the following pseudo code segment:
K: = 0 for **i1: = 1** to **n** for **i2 : = 1** to **i1** : For **im: =1**to **im – 1** **K: = K + 1** The value of K after the execution of this code shall be
- $C(n + m - 1, m)$
 - $C(n - m + 1, m)$
 - $C(n + m - 1, n)$
 - $C(n - 1 + 1, n)$

Ans. (a) : $C(n + m - 1, m)$

Because, In order to understand the solution to this problem let us consider some sample value of n and m . Let us say $n = 3$ and $m = 2$. The value of m also decides the number of inner loop we have. When we say $m = 2$, the total number of loops we have is 2 only n decides the range of outer most loop, so the pseudocode can be under stood as

```
k = 0
for i = 1 to n
  for m = 1 to i
    k = k + 1
```

$\Rightarrow$ For the value of $n = 3$ and $m = 2$, the value of k would be incriminated in following manner

i	m	k
1	1	1
2	1	2
2	2	3
3	1	4
3	2	5
3	3	6

$\Rightarrow$ Value of i ranges & rom 1 to n where $n = 3$

$\Rightarrow$ Value ko k is $k = 6$ at the end of iteration $C(n + m - 1/m) = C(3 + 2 - 2) = C(4, 2) = 4!/2! * 2! = 6.$

$\Rightarrow$ So the answer is A

2. **In Delta Rule for error minimization**

- Weights are adjusted w.r. To change in the output
- Weights are adjusted w.r. To difference between desired output and actual output
- Weights are adjusted w.r. To difference between input and output
- None of the above

Ans. (b) : in delta rule for error minimization, weights are adjusted with respect to difference between desired output and actual output.

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3. The concept of pipelining is most effective in improving performance if the tasks being performed in different stages:

- (a) require different amount of time
- (b) require about the same amount of time
- (c) require different amount of time with time difference between any two tasks being same
- (d) require different amount with time difference between any two tasks being different

Ans. (b) : Require about the same amount of time.

Because, Pipe lining is a technique to build fast processors it allows the execution of multiple instruction by overlapping them. Pipe lining can executes multiple instructions. So it requires the same amount of time. So that the answer is B.

4. What is Granularity?

- (a) The size of database
- (b) The size of data item
- (c) The size of record
- (d) The size of file

Ans. (b) : The size of data item is often called the data item granularity.

5. Suppose that given application is run on a 64-process or machine and that 70 percent of the application can be parallelized. Then the expected performance improvement using Amdahl's law is

- (a) 4.22
- (b) 3.22
- (c) 3.32
- (d) 3.52

Ans. (b) : 3.22

Because, According to Amdahl's law, in case of parallelization, if p is the proportion of the program that can be made parallel, then $(1-p)$ is the proportion can be paralyzed. Then the maximum speedup that can be achieved by using N processor, $S(N) = 1/(1-P) + P/N$, where N refers the number of processor and P refers to the proportion that can be paralyzed.

6. If two fuzzy sets A and B are given with membership functions $\mu_A(x) = \{0.2, 0.4, 0.8, 0.5, 0.1\}$ $\mu_B(x) = \{0.1, 0.3, 0.6, 0.3, 0.2\}$ Then the value of $\mu_{A \cap B}$ will be

- (a) $\{0.9, 0.7, 0.4, 0.8, 0.9\}$
- (b) $\{0.2, 0.4, 0.8, 0.5, 0.2\}$
- (c) $\{0.1, 0.3, 0.6, 0.3, 0.1\}$
- (d) $\{0.7, 0.3, 0.4, 0.2, 0.7\}$

Ans. (a) :

Give two fuzzy set

$$\mu_{\hat{A}} = \{0.2, 0.4, 0.8, 0.5, 0.1\}$$

$$\mu_{\hat{A}} = \{0.1, 0.3, 0.6, 0.3, 0.2\}$$

$$\text{so, } \mu_{\hat{A}} \cup_B(x) = \{0.2, 0.4, 0.8, 0.5, 0.2\}$$

and

$$\mu_{\hat{A}} \cup_B(x) = \{0.8, 0.6, 0.2, 0.5, 0.8\}$$

$$= 1 - \{0.1, 0.3, 0.6, 0.3, 0.1\}$$

$$= \{0.9, 0.7, 0.4, 0.7, 0.9\}$$

7. Match the following:

- | | |
|--------------------|-------------------|
| (a) OLAP | 1. Regression |
| (b) OLTP | 2. Data Warehouse |
| (c) Decision Tree | 3. RDBMS |
| (d) Neural Network | 4. Classification |
| (a) 2 3 1 4 | (b) 2 3 4 1 |
| (c) 3 2 1 4 | (d) 3 2 4 1 |

Ans. (b) : In OLTP database there is detailed and current data. In OLTP database schema used to store transaction data usually 3NF. OLAP is characterized by relatively low volume of transactions.

Decision tree is used for classification of data in data mining and AI.

Neural networks are used for the purpose of unsupervised learning, regression or classification.

8. Which level of Abstraction describes what data are stored in the Database?

- (a) Physical level
- (b) View level
- (c) Abstraction level
- (d) Logical level

Ans. (d) : Logical level

Because, Logical level describes what data are stored in the data base and what relationship among those data.

9. The problem that occurs when one transaction updates a database item and then the transaction fails for some reason is

- (a) Temporary Select Problem
- (b) Temporary Modify Problem
- (c) Dirty Read Problem
- (d) None

Ans (c) : The temporary (or dirty read) problem this problem occurs when one transaction updates a database item and then transaction fails for some reason. The updated item is accessed by another transaction before it is changed back to its original value.

10. In an image compression system 163^{84} bits are used to represent 256×256 image with 256 gray level. What is the compression ratio for this system?

- [illegible]

Ans. (*) number of bits required to store a 256×256 image with 256 gray level is

256 gray level is= 2^8

Therefore, compression ratio = $256 \times 256 \times 8 / 16384 = 32$

11. X.25 is Network.

- Connection Oriented Network
- Connection less Network
- Either Connection Oriented or Connection Less
- Neither Connection Oriented nor Connection Less

Ans. (a) X.25 is an ITU-T standard protocol suite for packet switched Wide Area Network communication.

NSAP addressing facility was added in the X.25 revision of the specification, and this enabled X.25 to better meet the requirements of OSI connection-Oriented Network service (CONS).

12. Which of the following can be used for clustering of data?

- (a) Single lever perception
- (b) Multilayer Perception
- (c) Self organizing map
- (d) Radial basis function

Ans. (c) : Self Organizing map is a type of Artificial Neural Network that is trained using unsupervised learning to produce a low dimensional maps. In maps consisting of thousands of nodes, it is possible to perform cluster operations on the map itself.

13. Which of the following is scheme to deal with deadlock?

- (a) Time out (b) Time in
(c) Both A & (B) (d) None of the above

Ans. (a) : Time out

Because, One of the strategy to avoid dead lock situation in java multithreading is using timeout. Suppose, one thread has acquired lock on one resource and now waiting for lock on another resource. After certain time period if it can not acquire lock on resource and then it should stop waiting for lock on resource 2. Also it should release lock on resource 1. Thus lock is avoided, so the Answer is a

14. It the pixels of an image are shuffled then the parameter that may change is

- (a) Histogram (b) Mean
(c) Entropy (d) Covariance

Ans. (d) : Covariance

Because, Covariance is a measure of how much two random variable change together. So when pixel of an image suffered then covariance is changed.

15. The common property of functional language and logical programming language

- (a) Both are declarative
- (b) both are based on 1-calculus
- (c) Both are procedural
- (d) Both are functional

Ans. (a) : Both are declarative

Functional and logical programming languages are characterized by declarative programming style. In logical programming languages, programs consist of logical statements and program executes by searching for proof of the statements.

16. Given the following statements:

(1) The power of deterministic finite State machine and nondeterministic finite state machine are same

(2) The power of deterministic pushdown automaton and nondeterministic pushdown automaton are same

Which of the above is the correct statement (s)?

- (a) Both 1 and 2
- (b) Only 1
- (c) Only 2
- (d) Neither 1 nor 2

Ans. (b) : Only (1)

Because, we are now that finite machine is of two types one is deterministic finite state machine and the other one non deterministic finite state machine. Both machine accept regular language only, so the power of DFA = NFA, so the first statement is true.

There is no algorithm exist which convert NPDA into DPDA. So, power of NDPA is more than DPDA.

hence the answer is (b)

17. Let $Q(x, y)$ denote " $x + y = 0$ " and let there be two quantifications given as

(i) $\exists y \forall x Q(x, y)$

(ii) $\forall x \exists y Q(x, y)$ which of the following is valid? where x & y are real numbers. Then

- (a) (i) is true & (ii) is false.
- (b) (i) is false & (ii) is true.
- (c) (i) is false & (ii) is also false.
- (d) both (i) & (ii) are true.

Ans. (b)

$\exists y \forall x Q(x + y = 0)$ is false

Since for all x single y not exist whose addition produce result 0.

i.e., $x + y = 0$

$\forall x \exists y Q(x + y = 0)$ is true

Because for every x then exist $y = -x$ whereas addition produce result as 0.

18. Consider a schema R (A, B, C, D) and functional dependencies $A \rightarrow B$ and $C \rightarrow D$. Then the decomposition $R_1 (A, B)$ and $R_2 (C, D)$ is
- (a) Dependency preserving but not lossless join
 - (b) Dependency preserving and lossless join
 - (c) Lossless join but not dependency preserving
 - (d) Lossless Join

Ans. (a) : Dependency preserving but not lossless join
Because, for lossless after decomposition of relation into if there is any common attribute in the decomposed relation and that attribute is key in any of relation then it is lossless.
Here no attribute are common R_1 and R_2
so this one is not lossless
For dependency from the dependencies of the spited schemas then it is dependency preserving
here from $R_1 \implies A \rightarrow B$ from $R_2 \implies C \rightarrow D$ can be divided so it is DP, then the answer is (a)

19. The quantiser in an image-compression system is a
- (a) lossy element which exploits the psycho visual redundancy
 - (b) lossless element which exploits the psycho visual redundancy
 - (c) lossy element which exploits the statistical redundancy
 - (d) lossless element which exploits the statistical redundancy

Ans. (a) : Lossy element which exploits the psycho visual redundancy
Because, Quantization, involved in image processing is a lossy compression technique achieved by compressing a range of values to single quantum value. When the number of discrete symbols in a given stream is reduced and the stream becomes more compressible.
so the answer is a

20. Data Warehouse provides
- (a) Transaction Responsiveness
 - (b) Storage. Functionality Responsiveness to queries
 - (c) Demand and Supply Responsiveness
 - (d) None of the above

Ans. (b) : Storage functionality responsiveness to queries
Because, data warehouses are supposed to provide storage functionality and responsiveness oriented databases. Also data warehouse are set to improve the data access performance of database.

21. A * algorithm $f = g + h$ to estimate the cost of getting from the initial state to the goal state, where g is a measure of the cost of getting from initial state to the current node and the function h is an estimate of the cost of getting from the current node to the goal state. To find a path involving the fewest number of steps, we should set

- (a) $g = 1$ (b) $g = 0$
(c) $h = 0$ (d) $h = 1$

Ans. (a) : $g = 1$

Because, a* algorithm is the most important form of best first search,

$F = g + h$

g is the measure of cost getting from initial to the current node and the function,

h is an estimate of the cost of getting from the current node to the goal state.

Now if want to find a path involving the first number of steps then we set the cost of going from a node to its successor (i.e. g) as a constant usually 1. hence the answer is a

22. The transform which possesses the highest 'energy compaction' property is

- (a) Slant transform
(b) Cosine transforms
(c) Fourier transforms
(d) Karhunen-Loeve transforms

Ans. (d) : Karhunen-Loeve transforms

Because, most of the signal information is concentrated in few low frequency components of the transform, so approaching the Karhunen-Loeve transforms, which is optimal in the decorrelation sense.

23. Which one of the following prolog programs correctly implement "if G succeeds then execute goal P else execute goal θ ?"

- (a) $\text{if-else } (G, P, \theta) :- !, \text{ call } (G), \text{ call } (P).$
 $\text{if-else } (G, P, \theta) :- \text{ call } (\theta)$
(b) $\text{if-else } (G, P, \theta) :- \text{ call } (G), !, \text{ call } (P).$
 $\text{if-else } (G, P, \theta) :- \text{ call } (\theta)$
(c) $\text{if-else } (G, P, \theta) :- \text{ call } (G), \text{ call } (P), !.$
 $\text{if-else } (G, P, \theta) :- \text{ call } (\theta)$
(d) All of the above

Ans. (b) : If G succeeds then execute goal P else execute goal θ :

If $\text{else } (G, P, \theta) : \text{ call } (G), !, \text{ call } (P)$

If $\text{else } (G, P, \theta) : \text{ call } (\theta)$

24. The memory allocation function modifies the previous allocated space.

- (a) $\text{cal loc } ()$ (b) $\text{free } ()$

(c) malloc ()

(d) real loc ()

Ans. (d) : real loc ()

Because, mal loc and cal loc are used to allocate dynamic memory.

free is used to free the allocated memory by cal loc and mal loc

Real loc is used to reallocate or modify the previous allocated space, hence the answer is d

25. Which is not the correct statement (s)?

- (a) Every context sensitive language is recursive.
- (b) There is a recursive language that is not context sensitive.
 - (i) 1 is true, 2 is false.
 - (ii) 1 is true and 2 is true.
 - (iii) 1 is false, 2 is false.
 - (iv) 1 is false and 2 is true.

Ans. (b) : Since context sensitive languages are subset of recursive languages.

Therefore, both statements are correct.

26. The mechanism that binds code and data together and keeps them secure from outside world is known as

- (a) Abstraction
- (b) Inheritance
- (c) Encapsulation
- (d) Polymorphism

Ans. (c) : Encapsulation is an OOP concept that binds together the data and functions that manipulate the data and that keeps both safe from outside interference and misuse. Data encapsulation led to the important OOP concept of data hiding.

27. Identify the addressing modes of below instructions and match them:

- (a) ADI 1. Immediate addressing
- (b) STA 2. Direct addressing
- (c) CMA 3. Implied addressing
- (d) SUB 4. Register addressing
 - (i) a – 1, b – 2, c – 3, d – 4
 - (ii) a – 2, b – 1, c – 4, d – 3
 - (iii) a – 3, b – 2, c – 1, d – 4
 - (iv) a – 4, b – 3, c – 2, d – 1

Ans. (a) : ADI. Immediate addressing

Because. the instruction ADI adds same content to the accumulator. It is an immediate addressing mode instruction. The instruction STA stores the content of the accumulator in the particular memory location specified as operand, CMA takes complement of the contents of the accumulator. SUB instruction subtracts the content of the register to the contents of the accumulator.

Hence the answer is a

28. Which one of the following is not a Greibach Normal form grammar?

(a) $S \rightarrow a|bA|aA|bB$

$A \rightarrow a$

$B \rightarrow b$

(b) $S \rightarrow a|aA|AB$

$A \rightarrow a$

$B \rightarrow b$

(c) $S \rightarrow a|A|aA$

$A \rightarrow a$

(a) 1 and 2

(b) 1 and 3

(c) 2 and 3

(d) 1, 2 and 3

Ans. (c) : (2) and (3)

Because, the symbol on the right hand side of the production must be terminal. It can be followed by zero or more variables in grammar (2) of the question $S \rightarrow ABv$ is the production. AB are two non-terminals can be in GNF (Greibach Normal Form Grammar). In grammar not allowed in GNF. So the grammar which is not in GNF is (2) and (3) so the option is c

29. Which of the following IP address class is a multicast address?

(a) Class A

(b) Class B

(c) Class C

(d) Class D

Ans. (d) : IPv4 multicast addresses are defined by the leading address bits of 1110, originating from the class full network design of the early internet when this group of addresses was designated as Class D.

30. While unit testing a module, it is found that for a set of test data, maximum 90% of the code alone were tested with a probability of success 0.9. The reliability of the module is

(a) at least greater than 0.9

(b) equal to 0.9

(c) at most 0.81

(d) at least 1/0.81

Ans. (c) : almost 0.81

Because, code tested maximum 90%

Probability of success is 0.9

so the reliability of module almost $0.9 \times 0.9 = 0.81$

hence the answer is option c

31. The upper bound computing time of m coloring decision problem is

(a) $O(nm)$

(b) $O(nm)$

(c) $O(nmn)$

(d) $O(nmmn)$

Ans. (c) : $O(nmn)$

Because, the $O(nm^n)$ is the only bound of computing time of M coloring.

32. The equivalent grammar corresponding to the grammar

**G : $S \rightarrow aA$,
 $A \rightarrow BB$,
 $B \rightarrow aBb | \epsilon$**

is

- (a) $S \rightarrow aA, A \rightarrow BB, B \rightarrow aBb$
- (b) $S \rightarrow a|aA, A \rightarrow BB, B \rightarrow aBb|ab$
- (c) $S \rightarrow a|aA, A \rightarrow BB|B, B \rightarrow aBb$
- (d) $S \rightarrow a|aA, A \rightarrow BB|B, B \rightarrow aBb|ab$

Ans. (d) : Given grammar is :

$S \rightarrow aA$
 $A \rightarrow BB$
 $B \rightarrow aBb | \epsilon$

Removing null-production from the grammar :
 Nullable variables are $\{A, B\}$, so,
 $S \rightarrow a|aA$
 $A \rightarrow B|BB$
 $B \rightarrow aBb|ab$

33. Which one of the following statements is incorrect?

- (a) The number of regions corresponds to the cyclomatic complexity.
- (b) Cyclometric complexity for a flow graph G is $V G = N - E + 2$, where E is the number of edges and N is the number of nodes in the flow graph.
- (c) Cyclometric complexity for a flow graph G is $V G = E - N + 2$, where E is the number of edges & N is the number of nodes in the flow graph.
- (d) Cyclometric complexity for a flow graph G is $VG = P + 1$, where P is the number of predicate nodes contained in the flow graph G.

Ans. (b) : Cyclometric complexity is a software metric (measurement) used to indicate the complexity of a program. The complexity M is defined as :

$M = E - N + 2P$

where, E is the number of edges, N is number of vertices and P is number of connected component of flow graph. For a single program (component) P is always equal to 1.

Also, the number of regions corresponds to the cyclomatic complexity (M) and

$M = P + 1$

where P (nodes whereas out degree more than 1) is number of predicated notes contained in the flow graph G.

34. Consider a weighted undirected graph with positive edge weights and let (u, v) be an edge in the graph. It is known that the shortest path from source vertex s to u has weight 53 and

shortest path from s to v has weight 65. Which statement is always true?

- (a) Weight (u, v) < 12
- (b) Weight (u, v) = 12
- (c) Weight (u, v) ≥ 12
- (d) Weight (u, v) > 12

Ans. (c) : Weight (u, v) ≥ 12

Because, if weight (u, v) < 12 then the min. weight of (s, u) = weight of (s, u) + weight of (u, v) = 53 + (<12) will be less than 65

so that answer is c

35. Consider the regular expression (a + b) ... (a + b) (n-times). The minimum number of states in finite automaton that recognizes the language represented by this regular expression contains

- (a) n states
- (b) n + 1 states
- (c) n + 2 states
- (d) 2n states

Ans. (b) : n + 1 states

Because, without trap for 1 symbol (a + b) or b, required 2 states and for 2 symbols (a + b) required 3 states and so on so, n + 1 states for NFA

n + 2 states for DFA (include trap state)

asking for minimum so n + 1 states

36. Number of binary trees formed with 5 nodes is

- (a) 32
- (b) 36
- (c) 120
- (d) 42

Ans. (d) : 42

Because, $C = \frac{{}^{2n}C_n}{n+1} = \frac{{}^{10}C_5}{6} = \frac{256}{6} = 42$

so the option is d

37. Are we building the right product? This statement refers to

- (a) Verification
- (b) Validation
- (c) Testing
- (d) Software quality assurance

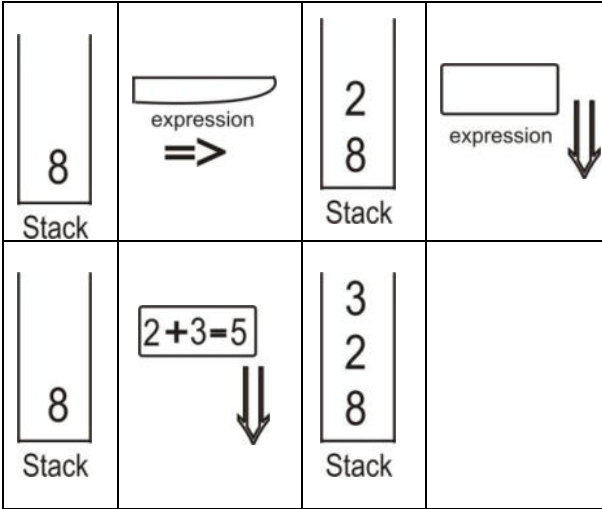
Ans. (b) :

- Validation : Are we building the right product?
- Verification : Are we building the product right?

38. The following postfix expression is evaluated using a stack $823^{\wedge}/23 * + 51 * -$ The top two elements of the stack after first * is evaluated

- (a) 6, 1
- (b) 5, 7
- (c) 3, 2
- (d) 1, 5

Ans. (a) : 6, 1
 Because, hence the answer is a (6, 1)



- 39. The following CFG $S \rightarrow aB \mid bA$, $A \rightarrow a \mid as \mid bAA$, $B \rightarrow b \mid bs \mid aBB$ Generates strings of terminals that have**
- (a) Odd number of a's and odd number of b's
 - (b) Even number of a's and even number of b's
 - (c) Equal number of a's and b's
 - (d) Not equal number of a's and b's

Ans. (c) : Given CFG is :
 $S \rightarrow aB \mid bA$
 $A \rightarrow a \mid aS \mid bAA$
 $B \rightarrow b \mid bS \mid aBB$
 The language of given CFG is set of all strings that have equal number of a's and b's. Other CFG may also possible for this language.

- 40. Consider the following pseudo-code: If (A > B) and (C > D) then**
A = A + 1
B = B + 1
Endif
The cyclomatic complexity of the pseudo-code is
- (a) 2
 - (b) 3
 - (c) 4
 - (d) 5

Ans. (b) : Cyclomatic complexity directly measures the number of linearly independent paths through a program's source code :
 Given, if (A > B) and (C > D) then
 A = A + 1
 B = B + 1
 end if
 Case (i) :
 if (true) and (true) then
 //execute some code
 end if
 Case (ii) :
 if (true) and (false) then
 //don't execute code
 end if

Case (iii) :

If (false) and (2^{nd} part won't be executed) then
//don't execute code
end if

So cyclomatic complexity will be 3.

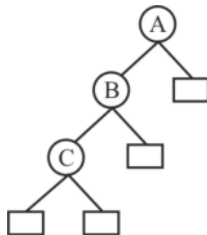
41. Which layer of OSI reference model uses the ICMP (Internet Control Message Protocol)?

- (a) Transport layer (b) Data link layer
(c) Network layer (d) Application layer

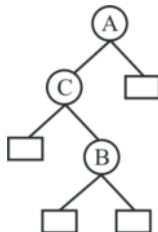
Ans. (c) : The internet Control Message Protocol (ICMP) is a network layer protocol that is used by network devices, like routers, to send error message and operational information indicating, for example, that a requested service is not available or that a host or router could not be reached.

42. Which one of the following binary search tree is optimal, if probabilities of successful search and unsuccessful search are same?

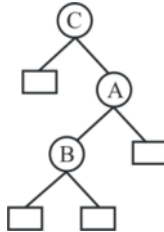
(a)



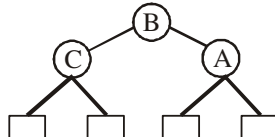
(b)



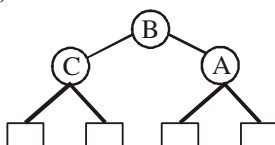
(c)



(d)

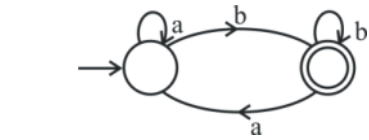


Ans. (d)



Given binary tree is balanced binary tree. So, it gives optimal.

43. The regular expression following DFA is



- (a) $ab^*(b + aa^*b)^*$
- (b) $a^*b(b + aa^*b)^*$
- (c) $a^*b(b^* + aa^*b)$
- (d) $a^*b(b^* + aa^*b)^*$

Ans. (b, d) Regular expression is

$$L = a^*b(b + aa^*b)^*$$

$$= a^*b(b^* + aa^*b)^*$$
 Both option (b) and (d) are true.

44. Which diagram provides a formal graphic notation for modelling objects, classes and their relationships to one another?

- (a) Object diagram
- (b) Class diagram
- (c) Instance diagram
- (d) Analysis diagram

Ans. (a) Object diagram provide a formal graphic notation for modeling objects, classes and their relationship to another. Object diagrams are useful for both abstract modeling and for designing actual program.

45. A computer system supports 32 bit virtual address as well as 32 bit physical addresses. Since the virtual address space is of same size as that of physical address space, if we want to get rid of virtual memory, which one of the following is true?

- (a) Efficient implementation of multiuser support is no longer possible.
- (b) The processor cache can be made more efficient.
- (c) Hardware support for memory management is not needed.
- (d) CPU scheduling can be made more efficient.

Ans. (c) : Virtual memory is a memory management technique that is implemented using both hardware and software. Virtual machine can employ hardware support to increase performance of their virtual memory implementations.

46. The feasible region represented by the constraints $x_1 - x_2 < 1$, $x_1 + x_2 \geq 3$, $x_1 \geq 0$, $x_2 \geq 0$ of the objective function $\text{Max } Z = 3x_1 + 2x_2$ is

- (a) A polygon
- (b) Unbounded feasible region
- (c) A point
- (d) None of these

$$\text{Max} = 3x_1 + 2x_2 \quad \dots(i)$$

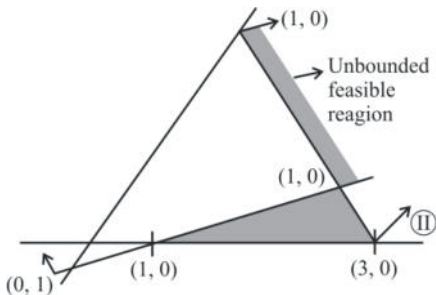
$$\text{Max} = 3x_1 + 2x_2 \quad \dots(i)$$

$$x_1 + x_2 = 1 \quad \dots(\text{ii})$$

$$x_1 + x_2 = 1 \quad \dots(\text{ii})$$

$$x_1 \geq 0, x_2 \geq 0; x_1 + x_2 \geq 3 \quad \text{.....(iii)}$$

$$x_1 \geq 0, x_2 \geq 0; x_1 + x_2 \geq 3 \quad \text{.....(iii)}$$



Feasible region is unbounded.

- 47. The colour of an object is largely determined by its diffuse reflection coefficient. If $K_d = (0.8, 0.4, 0)$, then what shall be the colour of the object, if the light used is blue and magenta?**
- (a) White and Red (b) Red and Blue
(c) Black and White (d) Black and Red

0.0 – for dull surface that absorbs almost all light.

1.0— for string surface that reflects almost all light.

For the value (0.8, 0.4, 0) the color of the object will be black and red.

48. If an instruction takes 'i' microseconds and a page fault takes an additional 'j' microseconds. The effective instruction time, if on the average a page fault occurs every k instructions, is
- (a) $1 + \frac{j}{k}$ (b) $1 + j * k$
- (c) $\frac{(i+j)}{k}$ (d) $(i + j) * k$

Ans. (a) : Given page fault rate $(1-p) = \frac{1}{K}$

Service time (m) = i

Page fault service time (S) = i + j

Therefore,

Effective memory access time is:

$$\text{EMAT} = i \cdot \left(1 - \frac{1}{k}\right) + (i + j) \times \frac{1}{k}$$

$$= \frac{k \cdot i - i}{k} + \frac{i + j}{k}$$

$$= \frac{ki - i + i + j}{k} = \frac{ki + j}{k}$$

$$\text{EMAT} = i + \frac{j}{k}$$

49. In any simplex table, if corresponding to any negative Δ_j , all elements of the column are negative or zero, the solution under the test is :
- (a) degenerate solution
 - (b) unbounded solution
 - (c) alternative solution
 - (d) non-existing solution

Ans. (b) : Procedure to test the basic feasible solution for optimality by the rules given :

Rule (i) : If all $\Delta_j \geq 0$, the solution under the test will be optimal. Alternate optimal solution will exist if any non-basic Δ_j is also zero.

Rule (ii) : If atleast one Δ_j is negative, the solution is not optimal and then proceeds to improve the solution in the next step.

Rule (iii) : If corresponding any negative Δ_j , all elements of the column x_j are negative or zero, then the solution under test will be unbounded. So, according to rule (iii) that is unbounded solution.

50. How many relations are there on a set with n elements that are symmetric and a set with n elements that are reflexive and symmetric?
- (a) $2^{n(n+1)/2}$ and $2^n \cdot 3^{n(n-1)/2}$
 - (b) $3^{n(n-1)/2}$ and $2^{n(n-1)}$
 - (c) $2^{n(n+1)/2}$ and $3^{n(n-1)/2}$
 - (d) $2^{n(n+1)/2}$ and $2^{n(n-1)/2}$

Ans. (d) : Symmetric relation : A relation R on set A is said to be symmetric if

(xRy) then (yRx) , $\forall x, y \in A$

Total number of symmetric relation with n element set

$$= 2^n \times 2^{\frac{n^2-n}{2}} = 2^{\frac{n(n+1)}{2}}$$

Total number of symmetric reflexive relation with n element set

$$= 1 \times 2^{\frac{n^2-n}{2}} = 2^{\frac{n(n-1)}{2}}$$

51. The strategy used to reduce the number of tree branches and the number of static evaluations applied in case of a game tree is
- (a) Minmax strategy
 - (b) Alpha-beta pruning strategy
 - (c) Constraint satisfaction strategy
 - (d) Static max strategy

Ans. (b) : Alpha-beta pruning is a search algorithm that seeks to decrease the number of nodes that are evaluated by the minimum algorithm in its search tree.

52. Match of following:

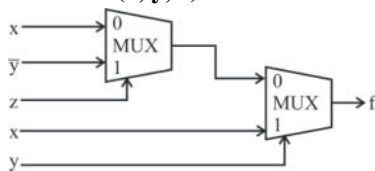
List-I		List-II	
A.	Regular Grammar	1.	Pushdown automaton
B.	Context free Grammar	2.	Linear bounded automaton
C.	Unrestricted Grammar	3.	Deterministic finite
D.	Context Sensitive Grammar	4.	Turning machine

Codes :

	A	B	C	D
(a)	3	1	2	4
(b)	3	1	4	2
(c)	3	2	1	4
(d)	3	2	4	1

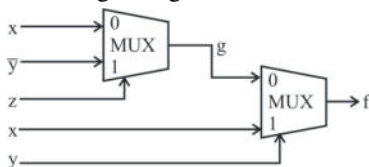
Ans. (b) :

53. Consider the below circuit and find the output function $f(x, y, z)$



- (a) $\bar{x}z + xy + \bar{y}z$ (b) $x\bar{z} + xy + \bar{y}z$
(c) $xz + xy + \bar{y}z$ (d) $xz + x\bar{y} + \bar{y}z$

Ans. (a) : Given logic diagram is :



Function f is :

$$\begin{aligned}
 &= y'(g) + y(x) \\
 &= y'(x.Z' + y'z) + x.y \\
 &= x.y'z' + y'z + x.y \\
 &= x(y + y'z') + y'z \\
 &= x(y + z') + y'z \\
 &= xy + x'z + y'z
 \end{aligned}$$

54. What is the size (in terms of bits) of Header length field in IPv4 header?

- (a) 2 (b) 4
(c) 8 (d) 16

Ans. (b) : Header length : The second field (4 bits) is the internet header length (IHL), which is the number of 32-bit words in the header. Since an IPv4 header may contain a variable number of options, this field specifies the size of the header (this is also coincides with the offset to the data). As a 4-bit field the minimum and maximum value of 5 and 15 respectively.

55. Match the following with respect to java. Util. * class methods:

List-I		List-II		
(a) Bit Set		(1) Time zone	getTime zone ()	
(b) Calendar		(2) int hashCode ()		
(c) Time zone		(3) int nextInt ()		
(d) Random		(4) Void setID (String tzName)		
	A	B	C	D
(a)	2	1	4	3
(b)	3	4	1	2
(c)	4	3	2	1
(d)	2	1	3	4

Ans. (a) :

56. is sometimes said to be object oriented, because the only way to manipulate kernel objects is by invoking methods on their handles.

(a) Windows NT (b) Windows XP
(c) Windows VISTA (d) Windows 95/98

Ans. (a) : Windows NT is sometimes said to object-oriented because the only way to manipulate Kernel objects is by invoking methods (API functions) on their handles. On the other hand, it lacks some of the most basic properties of object oriented systems such as inheritance and polymorphism.

57. A user level process in Unix traps the signal sent on a Ctrl + C input and has a signal handling routine that saves appropriate files before terminating the process. When a Ctrl + C input is given to this process, what is the mode in which the signal handling routine executes?

(a) User mode (b) Kernel mode
(c) Super user mode (d) Privileged mode

Ans. (b) : The Kernel handles signals in the context of the process that receives them, so a process must run to handle signals.

58. CPU generally handles an interrupt by executing an interrupt service routine

(a) As soon as an interrupt is raised
(b) By checking the interrupt register at the end of fetch cycle
(c) By checking the interrupt register after finishing the executing the current instruction
(d) By checking the interrupt register at fixed time intervals

Ans. (c) : After finishing the execution of each instruction the CPU reads the interrupt pins to recognize the interrupts.

59. The perspective projection matrix, on the view plane $z = d$ where the centre of projection is the origin $(0, 0, 0)$ shall be :

$$\begin{array}{ll}
 \text{(a)} \begin{bmatrix} 0 & 0 & 0 & d \\ 0 & 0 & d & 0 \\ 0 & d & 0 & 0 \\ d & 0 & 0 & 1 \end{bmatrix} & \text{(b)} \begin{bmatrix} d & 0 & 0 & 0 \\ 0 & d & 0 & 0 \\ 0 & 0 & d & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \\
 \text{(c)} \begin{bmatrix} 0 & 0 & 0 & d \\ 0 & 0 & d & 0 \\ 0 & d & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} & \text{(d)} \begin{bmatrix} d & 0 & 0 & 0 \\ 0 & d & 0 & 0 \\ 0 & 0 & d & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
 \end{array}$$

Ans. (b) : The projection transformation as a matrix

$$= \begin{bmatrix} d_0 & 0 & 0 & 0 \\ 0 & d_0 & 0 & 0 \\ 0 & 0 & d_0 & 0 \\ n_1 & n_2 & n_3 & 0 \end{bmatrix}$$

The plane $z = d$ is parallel, to the $x y$ plane ($N = K$)
 $N(n_1, n_2, n_3) = (0, 0, 1)$ and $R_0(x_0, y_0, z_0) = (0, 0, d)$
 So,

$$d_0 = n_1 x_0 + n_2 y_0 + n_3 z_0 = d$$

Therefore, the projection matrix is :

$$= \begin{bmatrix} d & 0 & 0 & 0 \\ 0 & d & 0 & 0 \\ 0 & 0 & d & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

60. Radio signals generally propagate according to the following mechanisms:

- Modulation, Amplification, Scattering
- Reflection, Diffraction, Scattering
- Amplification, Diffraction, modulation
- Reflection, Amplification, Diffraction

Ans. (b) : Reflection, diffraction and scattering are the three fundamental phenomenon that cause signal propagation in a mobile communication, apart from LOS communication.

61. Identify the devices given below with their IC numbers:

List-I		List-II		
(A) USART		(1) 8251		
(B) Micro controller		(2) 8051		
(C) Interrupt controller		(3) 8259		
(D) DMA controller		(4) 8257		
	A	B	C	D
(a)	1	2	3	4
(b)	2	1	4	3
(c)	3	4	1	2
(d)	4	1	2	3

Ans. (a) :

- A USART (Universal synchronous/ asynchronous receiver/transmitter) is manufactured by Intel with number as 8251A.
- A microcontroller is a small computer on a single integrated circuit. Small device C compiler as 8051.
- One of the best known programmable interrupt controller is 8259A, was included in the X86 PC.
- 8257 is known as microprocessor 8257 DMA controller.

62. The optimal solution of the following assignment problem using Hungarian method is

	(i)	(ii)	(iii)	(iv)
A	8	26	17	11
B	13	28	4	26
C	38	19	18	15
D	19	26	24	10

Codes :

- | | A | B | C | D |
|-----|----------|----------|----------|----------|
| (a) | (i) | (ii) | (iii) | (iv) |
| (b) | (i) | (iii) | (ii) | (iv) |
| (c) | (i) | (iii) | (iv) | (ii) |
| (d) | (i) | (iv) | (ii) | (iii) |

Ans. (b) :

- (a) Cost = $8 + 28 + 18 + 10 = 64$
- (b) Cost = $8 + 4 + 19 + 10 = 41$
- (c) Cost = $8 + 4 + 15 + 26 = 53$
- (d) Cost = $8 + 26 + 19 + 24 = 77$

63. If a and b are the end points of a line, then which one of the following is true?

- (a) If both end points are left, right, above or below the window, the line is invisible.
- (b) If both end points are left, right, above or below the window, the line is completely visible.
- (c) If both end points are left, right, above or below, the line is trivially visible.
- (d) If both end point are left, right, above or below the window, the line is trivially invisible.

Ans. (d) : In simple visibility algorithm, if a and b are the end points of the line, with components x and y for each line. Visibility is true then check for totally invisible lines. If both endpoints are left, right, above or below the window, then line is trivially invisible.

64. Identify the devices given below with their IC numbers :

List-I	List-II
(A) Networked-Controlled Handoff (NCHO)	(1) MS connect to BS
(B) Mobile-Assisted Handoff (MAHO)	(2) Process via channel the target BS
(C) Forward Handoff	(3) First Generation Analog Cellular System
(D) Hard Handoff	(4) Second Generation Digital Cellular System

	A	B	C	D
(a)	3	4	2	1
(b)	2	3	1	4
(c)	2	1	4	3
(d)	4	3	1	2

Ans. (a) :

65. Consider the methods used by processes P_1 and P_2 for accessing their critical sections. The initial values of shared Boolean variables S_1 and S_2 are randomly assigned,

P_1	P_2
While ($S_1 = S_2$); critical section $S_1 = S_2$;	While ($S_1 = S_2$); critical section $S_1 = S_2$;

Which one of the following statements describes the properties achieved?

- (a) Mutual exclusion but not progress
- (b) Progress but not mutual exclusion
- (c) Neither mutual exclusion nor progress
- (d) Both mutual exclusion and progress

Ans. (c) : On first attempt both process can go into critical section simultaneously. So mutual exclusion is not satisfied.

After first attempt neither process can go into critical section. So progress is also not satisfied.

66. If the period of a signal is 1000 ms, then what is its frequency in kilohertz?

- (a) 10^{-3} KHz
- (b) 10^{-2} KHz
- (c) 10^{-1} KHz
- (d) 1 KHz

Ans. (a) : Frequency = $\frac{1}{1000\text{msec}} = 10^{-3}$ KHz

67. Let $a * H$ and $b * H$ be two co-sets of H .

- (i) Either $a * H$ and $b * H$ are disjoint
- (ii) $a * H$ and $b * H$ are identical

Then

- (a) only (i) is true
- (b) only (ii) is true
- (c) (i) or (ii) is true
- (d) (i) and (ii) is false

Ans. (c) : Theorem : Any two right (or left) co-sets of H are either disjoint or identical.

68. HTML is defined using SGML an standard, information processing-text and office systems (SGML) for text information processing.

- (a) ISO – 8878
- (b) ISO – 8879
- (c) ISO – 8880
- (d) ISO – 8881

Ans. (b) : The Standard Generalized Markup Language (SGML; ISO 8879 : 1986) is a standard for defining generalized markup language for documents.

69. What is the meaning of 'Hibernate' in Windows XP/Windows 7?

- (a) Restart the computers in safe mode.
- (b) Restart the computers in normal mode.
- (c) Shutdown the computer terminating all the running applications.
- (d) Shutdown the computer without closing the running applications.

Ans. (d) : Hibernation (or suspend to disk) in computing is powering down a computer while retaining its state. Upon hibernation the computer saves the contents its Random Access Memory (RAM) to a hard disk or other non-volatile storage.

70. Assume that we have constructor functions for both base class and derived class. Now consider the declaration in main(). Base * P = New Derived: in what sequence will the constructor be called?

- (a) Derived class constructor followed by Base class constructor
- (b) Base class constructor followed by derived class constructor.
- (c) Base class constructor will not be called.
- (d) Derived class constructor will not be called.

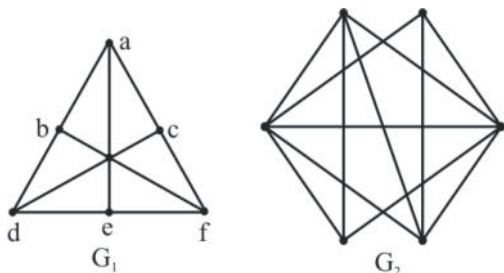
Ans. (b) : Here A derived class object is created first and then assigned to a base class pointer, whenever a derived class object is created, the base class constructor gets called first and then the derived class constructor.

71. Which one of the following options is not a shell in UNIX system?

- (a) Bourne Shell (b) C Shell
(c) Net Shell (d) Korn Shell

Ans. (c) : Net shell is a command line-based tool that enables administrators to remotely administer and configure critical network service. Net shell is not a shell of UNIX shell.

72. G_1 and G_2 are two graphs as shown :



- (a) Both G_1 and G_2 are planar graphs.
(b) Both G_1 and G_2 are not planar graphs.
(c) G_1 is planar and G_2 is not planar graph.
(d) G_1 is not planar and G_2 is planar graph

Ans. (d) : In planar graph :

(i) $e \leq 2n - 4$, (ii) $e \leq 3n - 6$ if Δ

$G_1 : n = 6, e = 9$

$$\Rightarrow e \leq 2n - 4$$

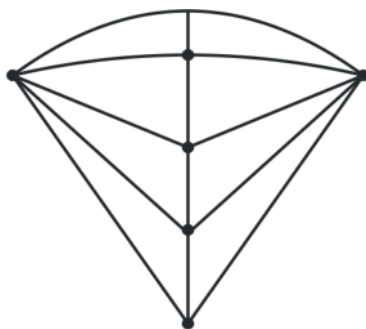
$$\Rightarrow 9 \leq 2 \times 6 - 4$$

$$\Rightarrow 9 \leq 8 \text{ false}$$

$G_2 : N = 6, e = 11$

$$\Rightarrow 11 \leq 3 \times 6 - 6$$

$$\Rightarrow 11 \leq 12 \text{ true}$$



Therefore, only G_2 is planar graph.

73. In which file the compiler manage the various objects, which are used in windows programming?

- (a) Control File (b) Binary File
(c) Text File (d) Obj File

Ans. (c) :

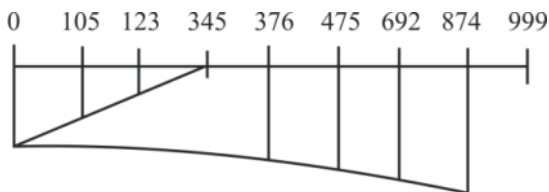
- Text files are stored in a form (usually ASCII) that is human readable.
- Control file is a small binary file that records the physical structure of the database.
- A binary file is computer readable but not human readable.
- Object file, and organized machine code file created by a compiler.

Microsoft windows computer text files are identified with the .txt file extension.

74. On a disk with 1000 cylinders (0 to 999) find the number of tracks, the disk arm must move to satisfy all the requests in the disk queue. Assume the last request service was at track 345 and the head is moving toward track 0. The queue in FIFO order contains requests for the following tracks; 123, 874, 692, 475, 105, 376 (Assume SCAN algorithm)

- (a) 2013 (b) 1219
(c) 1967 (d) 1507

Ans. (b) : Scan algorithm :



Total path = $(345 - 0) + (874 - 0) = 1219$

75. Half toning is defined as:

- (a) A technique to obtain increased visual resolution using multiple intensity levels.
- (b) A technique for using minimum number of intensity levels to obtain increased visual resolution.
- (c) A technique to obtain increased visual resolution using maximum number of intensity levels.
- (d) A technique for using appropriate number intensity levels to obtain increased visual resolution.

Ans. (b) : Antialiasing is a technique using multiple intensity levels to obtain increased visual resolution. Halftoning on the other hand, is a technique for using a minimum number of intensity levels, i.e. gray scaling or multiple intensity levels.